De La Salle University

SEMPHY01 — Semiconductor Physics

Long Assessment 1

From Bloch Bands to Equilibrium Semiconductor Carrier Statistics

Student:	TODO: Student Name
Student ID:	TODO: Student ID
Selected material:	TODO: Material and phase/polytype
Submission date:	TODO: Submission date
Computational record:	https://example.invalid/replace-me
Local module revision:	TODO: local module file hashes or revision

This report concerns equilibrium carrier statistics only. Transport, mobility, conductivity, device behavior, and spatial band bending are outside the model unless explicitly identified as excluded context.

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1 Project declaration and scope

Table 1: Declared scope of the material model.

Item	Declaration
Material identity	**TODO:** Chemical formula and exact phase/polytype.
Sample model	**TODO:** Ideal bulk model and temperature range.
Gap model	**TODO:** Direct/indirect character and $E_g(T)$ representation.
Conduction edge	**TODO:** Location, equivalent valleys, and mass model.
Valence edge	**TODO:** Relevant bands, degeneracies, and mass model.
Statistics	**TODO:** Full Fermi–Dirac, Boltzmann comparison, or both.
n-type model	**TODO:** Donor-like species/defect and charge states.
p-type model	**TODO:** Acceptor-like species/defect and charge states.
Exclusions	**TODO:** Omitted physics and why it is outside scope.

1.1 Research question and modeling strategy

TODO: State the material-specific question investigated and summarize the path from the Bloch calculation to intrinsic and doped equilibrium statistics.

1.2 Use of local libraries and student-authored work

The assessment supplies inspectable modules colocated with the notebooks. Distinguish supplied local capabilities from your own configuration, analysis, extension, and interpretation. The submitted calculation must not depend on Project Koios or PhysKit. Compare the adopted carrier-statistics conventions with an established semiconductor reference such as Sze and Ng [1].

Table 2: Computational capability and contribution map.

Capability	Source	Configuration, extension, or validation performed
Bloch Hamiltonian	`notebooks/bloch2d.py`	**TODO:** Revision, model, parameters, and checks.
DOS/state counting	**TODO:** Source	**TODO:** Normalization and validation.
Carrier statistics	**TODO:** Source	**TODO:** Integration/root-solving choices.
Dopant model	**TODO:** Source	**TODO:** Charge-state convention and extension.
Visualization	**TODO:** Source	**TODO:** How plots consume calculated results.

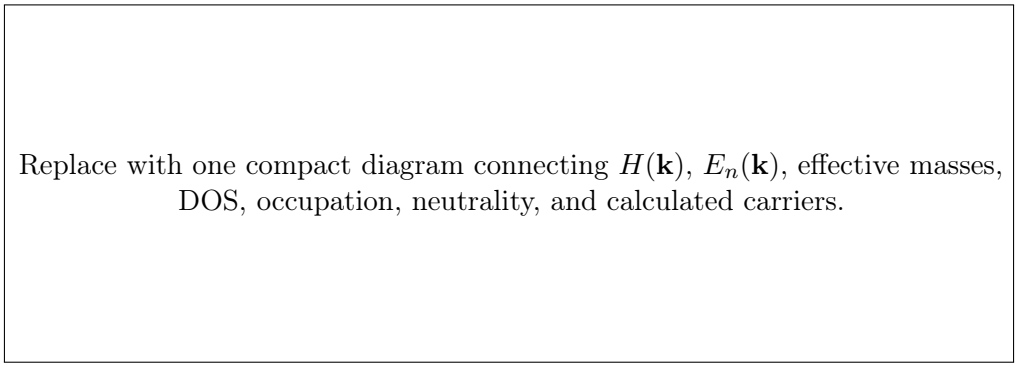


Figure 1: Replace with one compact diagram connecting $H(\mathbf{k})$, $E_n(\mathbf{k})$, effective masses, DOS, occupation, neutrality, and calculated carriers.

1.3 Model map

2 Part I: Bloch2D effective-mass reduction

2.1 Parent model and numerical representation

TODO: Describe the direct/reciprocal lattice, cell, boundary conditions, potential, representation, reciprocal cutoff or discretization, and units. Explain what the local plane-wave module supplies and what you configured.

A useful operator statement is

$$H_{\mathbf{G}\mathbf{G}'}(\mathbf{k}) = \frac{\hbar^2|\mathbf{k}+\mathbf{G}|^2}{2m}\delta_{\mathbf{G}\mathbf{G}'} + V_{\mathbf{G}-\mathbf{G}'}. \tag{1}$$

TODO: Report a Hermiticity or equivalent operator-consistency check.

2.2 Band path and full-zone calculation

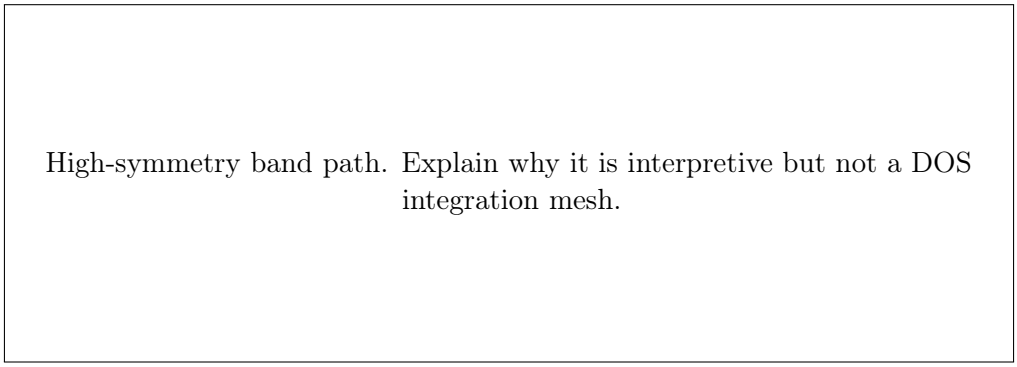


Figure 2: High-symmetry band path. Explain why it is interpretive but not a DOS integration mesh.

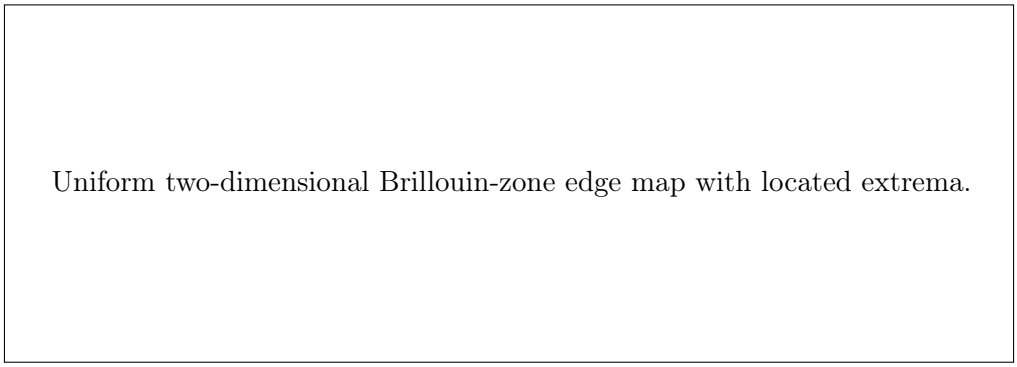


Figure 3: Uniform two-dimensional Brillouin-zone edge map with located extrema.

Table 3: Located band edges and fit domains.

Edge	Band/sheet	k_x	k_y	Energy	Classification
Valence maximum	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**
Conduction minimum	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**

2.3 Band edges and local quadratic model

For a conduction minimum,

$$E_c(\mathbf{k}_c + \mathbf{q}) \simeq E_{\mathrm{C}} + \frac{\hbar^2}{2}\mathbf{q}^{\mathsf{T}}\mathbf{M}_c^{-1}\mathbf{q}, \tag{2}$$

with the sign reversed before defining a positive hole-mass tensor at a valence maximum.

TODO: Describe the multidimensional fit, energy/radius window, tensor components, principal masses, principal directions, and any degeneracy or band folding that complicates the fit.

Table 4: Effective-mass tensor results.

Edge	$(M^{-1})_{xx}$	$(M^{-1})_{xy}$	$(M^{-1})_{yy}$	m_1/m_0	m_2/m_0	Fit window
Valence	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**
Conduction	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**

2.4 Convergence, residuals, and range of validity

TODO: Separate representation error, **k**-mesh error, fit-window error, and physical nonparabolicity. State a bounded energy range over which the parabolic approximation is adequate.

2.5 Two-dimensional DOS and state counting

For one anisotropic parabolic valley,

$$m_{d,2D} = \sqrt{m_1 m_2}, \qquad g_{2D}(E) = \frac{g_s g_v m_{d,2D}}{2\pi\hbar^2}\Theta(E - E_{\mathrm{C}}). \tag{3}$$

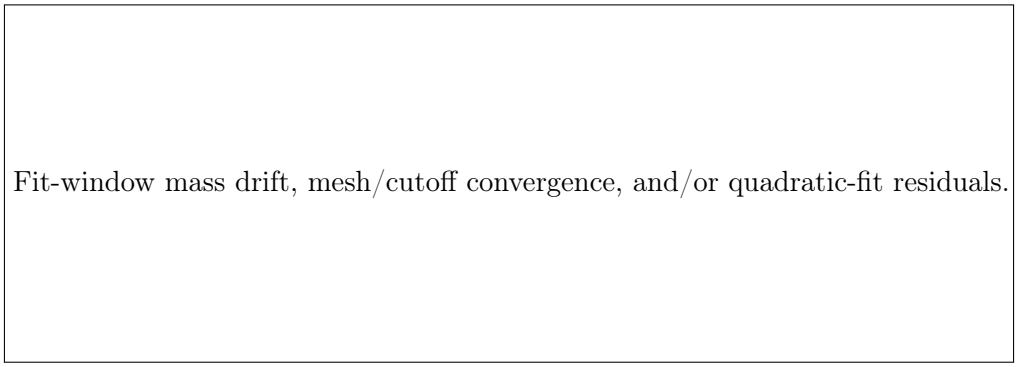


Figure 4: Fit-window mass drift, mesh/cutoff convergence, and/or quadratic-fit residuals.

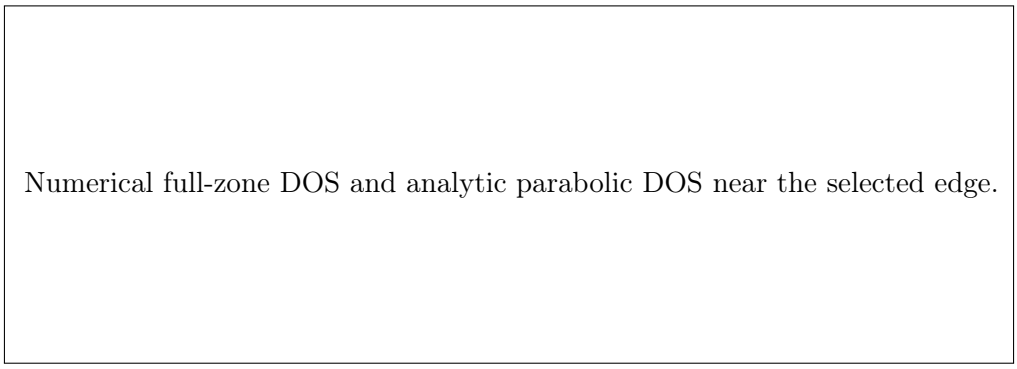


Figure 5: Numerical full-zone DOS and analytic parabolic DOS near the selected edge.

TODO: Verify integrated state counts and discuss mesh spacing, broadening, finite bandwidth, additional valleys/sheets, and nonparabolicity. Emphasize that these are two-dimensional sheet quantities.

3 Part II: Intrinsic three-dimensional model

3.1 Band-edge state model and DOS components

TODO: Define the energy reference, gap, all conduction valleys and valence bands, principal masses, and degeneracies. Explain whether quoted DOS masses already include valley multiplicity.

For each three-dimensional parabolic component,

$$g_c(E) = \frac{g_s g_v}{4\pi^2}\left(\frac{2}{\hbar^2}\right)^{3/2}\sqrt{\det \mathbf{M}_c}\sqrt{E - E_{\mathrm{C}}}\,\Theta(E - E_{\mathrm{C}}), \tag{4}$$

with an analogous expression below E_{V} for holes.

Table 5: Components of the effective densities of states at a representative temperature.

Component	g_s	g_v	m_1/m_0	m_2/m_0	m_3/m_0	Contribution
TODO: Conduction valley	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**
TODO: Valence band	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**

3.2 Fermi–Dirac occupation and carrier integrals

$$f(E,\mu,T) = \frac{1}{1+\exp[(E-\mu)/(k_{\mathrm{B}}T)]}, \tag{5}$$

$$n(\mu,T) = \int_{E_{\mathrm{C}}}^{\infty} g_c(E) f(E,\mu,T)\, dE, \tag{6}$$

$$p(\mu,T) = \int_{-\infty}^{E_{\mathrm{V}}} g_v(E)[1-f(E,\mu,T)]\, dE. \tag{7}$$

TODO: Explain stable evaluation, units, integration method, cutoffs, tolerances, and convergence evidence.

3.3 Intrinsic neutrality and nondegenerate comparison

$$n(\mu_{\mathrm{i}},T) - p(\mu_{\mathrm{i}},T) = 0. \tag{8}$$

TODO: Report the root bracket, tolerance, residual, and temperature sweep.

$$n_{\mathrm{i}} = \sqrt{N_{\mathrm{C}}N_{\mathrm{V}}}\exp\left(-\frac{E_{\mathrm{g}}}{2k_{\mathrm{B}}T}\right), \tag{9}$$

$$\mu_{\mathrm{i}} = \frac{E_{\mathrm{C}}+E_{\mathrm{V}}}{2} + \frac{k_{\mathrm{B}}T}{2}\ln\left(\frac{N_{\mathrm{V}}}{N_{\mathrm{C}}}\right). \tag{10}$$

Three-dimensional DOS and occupation at one temperature.

Figure 6: Three-dimensional DOS and occupation at one temperature.

TODO: Compare full Fermi–Dirac integration with the nondegenerate limit and identify when they agree.

3.4 Silicon control and validation evidence

TODO: Summarize automated or reproducible checks for monotonicity, energy-shift invariance, isotropic/anisotropic limits, $T^{3/2}$ scaling, neutrality, and integration convergence.

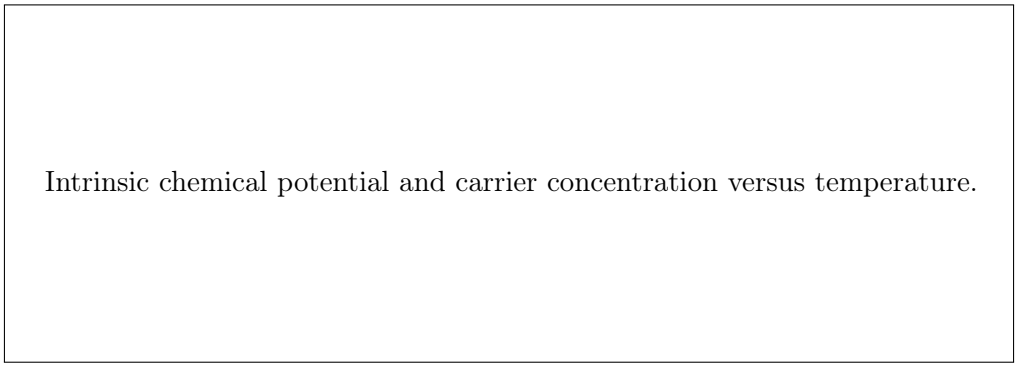


Figure 7: Intrinsic chemical potential and carrier concentration versus temperature.

Table 6: Silicon control at $3.00 \times 10^2\,\mathrm{K}$.

Quantity	Input/result	Reference value	Difference	Interpretation
E_g	**TODO:**	approximately 1.12 eV	**TODO:**	**TODO:**
n_i	**TODO:**	approximately $1.5 \times 10^{10}\,1/\mathrm{cm}^3$	**TODO:**	**TODO:**
μ_i	**TODO:**	**TODO:** source-dependent	**TODO:**	**TODO:**

4 Part III: Literature-based material reconstruction

4.1 Source strategy and parameter provenance

TODO: Explain how curated sources, primary literature, reviews, and published band information were reconciled. Do not mix phases or definitions silently.

Table 7: Material-parameter provenance.

Quantity	Symbol	Value/model	Unit	Temperature	Definition	Source
Band gap	E_g	**TODO:**	eV	**TODO:**	direct/indirect	**TODO:** cite
Electron mass	$\mathbf{M}_c$	**TODO:**	m_0	**TODO:**	curvature/DOS	**TODO:** cite
Hole mass	$\mathbf{M}_v$	**TODO:**	m_0	**TODO:**	HH/LH/DOS	**TODO:** cite
Valley count	g_v	**TODO:**	1	**TODO:**	included in mass?	**TODO:** cite
Published N_C	N_C	**TODO:**	cm^{-3}	**TODO:**	measured/derived	**TODO:** cite
Published N_V	N_V	**TODO:**	cm^{-3}	**TODO:**	measured/derived	**TODO:** cite
Published n_i	n_i	**TODO:**	cm^{-3}	**TODO:**	method	**TODO:** cite

4.2 Band-diagram interpretation

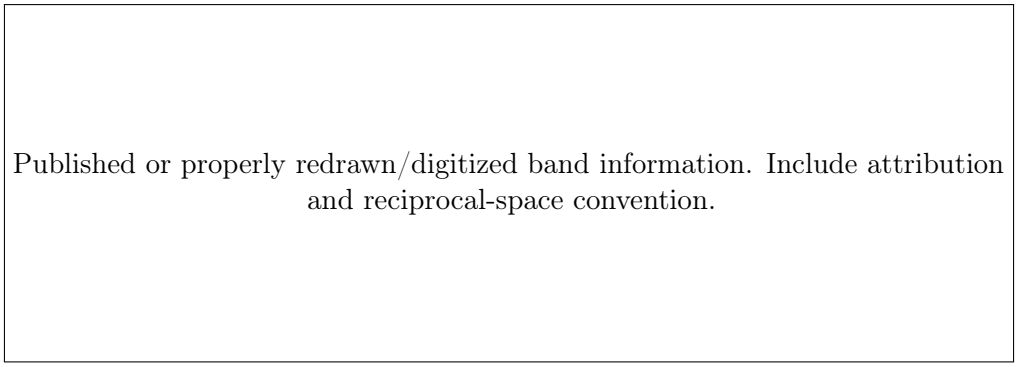


Figure 8: Published or properly redrawn/digitized band information. Include attribution and reciprocal-space convention.

TODO: Identify edge locations, direct/indirect character, multiplicity, nearby bands, anisotropy, nonparabolicity, and limitations of the plotted path.

4.3 Material model and intrinsic results

TODO: Define $E_g(T)$, modeled components, temperature range, and any material-specific use or extension of the colocated semiconductor module.

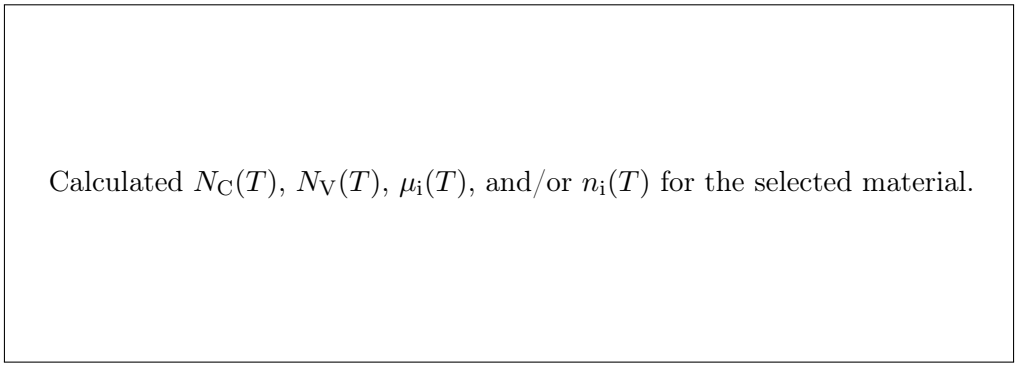


Figure 9: Calculated $N_C(T)$, $N_V(T)$, $\mu_i(T)$, and/or $n_i(T)$ for the selected material.

4.4 Published comparison and sensitivity

Table 8: Comparison with independent published data.

Quantity	T	Calculated	Published	Unit	Difference	Source
TODO:	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:** cite

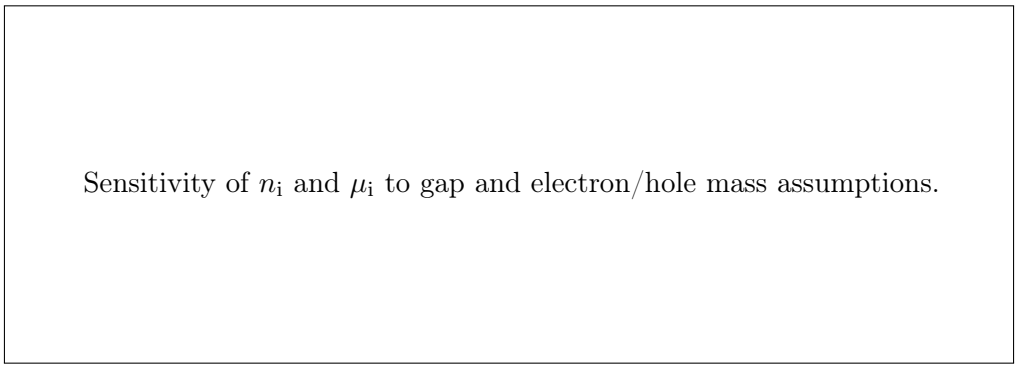


Figure 10: Sensitivity of n_i and μ_i to gap and electron/hole mass assumptions.

TODO: Separate numerical error, parameter uncertainty, and physical model limitations when interpreting disagreement.

Table 9: Dopant or defect-level provenance.

Kind	Species/site	Level	Charge states	Degeneracy convention	Range	Source/limitations
Donor	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:** cite
Acceptor	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:**	**TODO:** cite

5 Part IV: Equilibrium doping

5.1 Donor and acceptor models

With the explicitly declared degeneracy convention, one possible model is

$$N_D^+ = \frac{N_D}{1 + g_D \exp[(\mu - E_D)/(k_\mathrm{B}T)]}, \tag{11}$$

$$N_A^- = \frac{N_A}{1 + g_A \exp[(E_A - \mu)/(k_\mathrm{B}T)]}, \tag{12}$$

$$0 = p + N_D^+ - n - N_A^-. \tag{13}$$

TODO: Explain the convention, low/high-temperature behavior, solver bracket, and neutrality residual.

5.2 Representative intrinsic, n-type, p-type, and compensated cases

Table 10: Modeled equilibrium cases.

Case	N_D	N_A	Temperature range	Physical purpose
Intrinsic	0	0	**TODO:**	baseline
n-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**
compensated n-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**
p-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**
compensated p-type	**TODO:**	**TODO:**	**TODO:**	**TODO:**

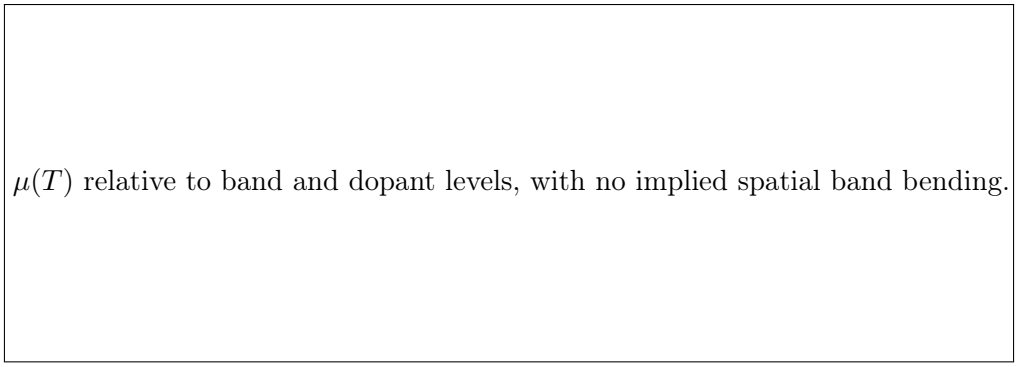


Figure 11: $\mu(T)$ relative to band and dopant levels, with no implied spatial band bending.

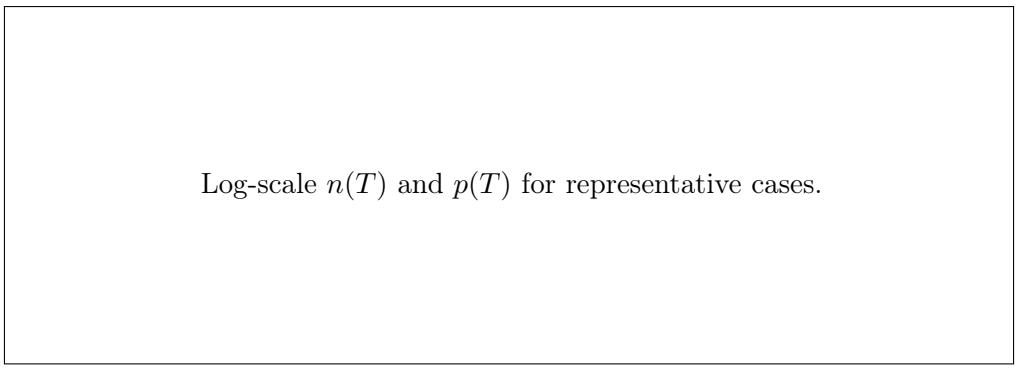


Figure 12: Log-scale $n(T)$ and $p(T)$ for representative cases.

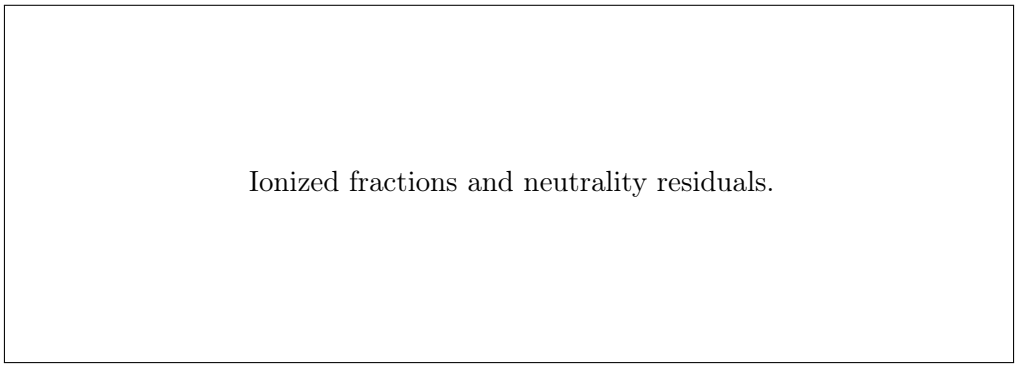


Figure 13: Ionized fractions and neutrality residuals.

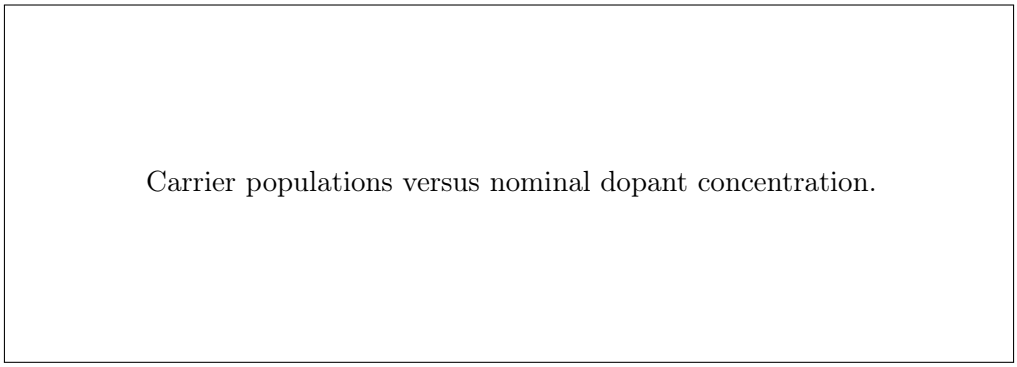


Figure 14: Carrier populations versus nominal dopant concentration.

5.3 Approximation comparison and regimes

TODO: Identify freeze-out, extrinsic, intrinsic, and degenerate/non-Boltzmann regimes where physically accessible. Compare incomplete and full ionization, and state where the simpler result differs by less than five percent.

6 Synthesis and bounded validity

6.1 What transfers from Bloch2D to the real material?

TODO: Separate transferable numerical ideas—curvature fitting, state counting, occupation, convergence—from dimensionality and material-specific physics that do not transfer directly.

6.2 Adequacy of the final model

TODO: Answer quantitatively: over what temperature and concentration range is the effective-mass plus equilibrium dopant model adequate, and which omitted physical effect is most likely to invalidate it first?

6.3 Main conclusion

TODO: Give a concise answer supported by your most decisive table and figure.

7 Reproducibility and validation record

Table 11: Computational environment and execution record.

Item	Value
Repository/archive	`https://example.invalid/replace-me`
Python version	**TODO:**
Local module revision	TODO: local module file hashes or revision
Main dependencies	**TODO:**
Run order	**TODO:**
Random seeds, if any	**TODO:**
Clean-run result	**TODO:**
Test/validation summary	**TODO:**

TODO: State how the submitted figures and tables can be regenerated. Disclose materially reused code, data, and computational assistance according to course policy.

References

[1] S. M. Sze and Kwok K. Ng. *Physics of Semiconductor Devices*. Wiley, 3 edition, 2007.

A Supplementary convergence and validation evidence

TODO: Place useful evidence here when it would interrupt the main argument. Do not use the appendix to hide checks needed to support a main conclusion.

B Additional parameter and provenance records

TODO: Include extended tables, digitization details, or source-definition comparisons if needed.